

Analysis

1. Load in required libraries
2. Load in processed dataset (bus_stops_clean)
3. Perform Chi-squared test: Is there a statistically significant difference in infrastructure between RideOn and Metro bus stops?
 - a. All amenities (benches, shelters and trashcans) have significant p-values even after Bonferroni correction. This means that the type of route (Metro vs. RideOn) is strongly associated with whether a stop has infrastructure which reflects what was seen in the visualizations.
 - b. While we know there is an association, chi-squared tests don't indicate why.
4. Nearest Neighbors (NN) analysis
 - a. Find average distance between bus stops with benches in two towns with the lowest bench rates (Olney and Clarksburg)
 - i. Both towns have very large, unwalkable average distances between their bus stops with benches.
 - b. Scatter plot of 'Bench Rate vs. Nearest Neighbors distance'
 - i. Towns with lower bench rates tend to have larger distances between bench-equipped stops.
 - ii. Clarksburg is an extreme outlier (very low bench rate and very high NN distance)
 - c. Leaflet map of Clarksburg bus stops
 - i. The two bus stops that the town does have are positioned in front of a high school and the Montgomery County Correctional Facility which suggests thoughtful placement based on ridership volume.